

NORTH SEA STATION 63103, United Kingdom

WMO: 995780

Lat: 61.100N

Lon: 1.100E

Elev: 0

StdP: 101.33

Time Zone: 0.00 (EUW)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.4	1.3	-7.9	1.9	2.4	-6.7	2.1	3.1	22.6	4.3	21.6	4.5	8.8	350	0.921

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.9	17.4	15.3	16.5	14.8	15.6	14.0	16.0	17.1	15.3	16.3	14.6	15.5	6.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.0	16.6	14.8	10.5	15.9	14.0	10.0	15.1	44.8	17.2	42.9	16.5	40.8	15.6	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	18.1	15.9	DB	-1.4	19.6	1.0	2.4	-2.1	21.3	-2.7	22.7	-3.3	24.0	-4.0	25.7	(4)
(5)			WB	-2.6	17.0	1.2	1.8	-3.4	18.3	-4.1	19.4	-4.8	20.4	-5.6	21.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	5.9	5.3	6.0	6.9	8.8	10.8	13.2	13.7	12.4	9.5	8.2	6.2	
	DBStd	3.39	1.75	2.06	1.96	1.70	1.52	1.73	1.62	1.54	1.70	1.74	1.62	2.32	
	HDD10.0	728	126	131	124	92	43	8	0	0	3	28	54	118	
	HDD18.3	3429	384	364	383	342	296	226	159	143	179	273	303	376	
	CDD10.0	341	0	0	0	1	5	32	99	116	73	14	2	0	
	CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	8.3	10.1	9.8	9.3	8.0	6.9	6.6	6.5	6.6	7.9	9.0	9.7	9.7	
Precipitation	PrecAvg	1341	161	132	127	72	48	65	81	100	117	141	139	156	
	PrecMax	1652	252	257	237	123	83	132	153	162	190	216	220	240	
	PrecMin	954	76	58	46	26	22	26	40	66	48	58	43	65	
	PrecStd	165	48	53	46	23	16	24	22	25	34	41	39	42	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	10.0	9.7	11.2	13.0	18.1	18.8	19.6	17.1	13.9	11.6	10.7	
		MCWB	8.6	8.7	8.2	9.3	9.3	14.2	16.3	16.5	14.9	12.1	10.1	9.4	
	2%	DB	9.4	9.2	9.2	10.1	12.0	15.7	17.4	17.2	15.6	13.2	11.1	10.2	
		MCWB	7.8	7.8	7.9	8.6	9.6	13.5	15.7	15.4	14.2	11.5	9.8	8.8	
	5%	DB	8.9	8.8	8.9	9.5	11.4	13.7	16.4	16.5	15.0	12.5	10.6	9.9	
		MCWB	7.2	7.3	7.6	8.2	9.4	12.2	15.1	14.7	13.8	10.8	9.3	8.4	
	10%	DB	8.5	8.4	8.5	9.1	10.7	12.7	15.4	15.7	14.5	11.7	10.3	9.4	
		MCWB	6.8	6.8	7.1	7.8	9.0	11.2	14.2	13.9	13.2	10.0	8.8	7.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.3	9.2	10.4	11.1	15.2	17.2	16.8	15.6	13.1	10.9	10.2
			MCDB	9.6	9.5	9.4	10.4	11.7	17.1	18.0	18.1	16.4	13.7	11.3	10.4
2%		WB	8.5	8.7	8.6	9.2	10.5	13.8	16.2	16.0	14.8	12.1	10.4	9.6	
		MCDB	9.2	9.1	9.0	9.5	11.3	16.0	17.0	17.2	15.6	13.1	10.9	10.1	
5%		WB	7.9	8.2	8.1	8.8	10.0	12.9	15.5	15.3	14.2	11.3	9.9	8.8	
		MCDB	8.7	8.7	8.6	9.2	10.8	14.5	16.4	16.4	14.8	12.2	10.5	9.6	
10%		WB	7.2	7.4	7.6	8.2	9.5	12.0	14.6	14.5	13.8	10.6	9.3	8.0	
		MCDB	8.3	8.2	8.3	8.9	10.5	12.8	15.5	15.5	14.4	11.6	10.1	9.1	
Mean Daily Temperature Range		5% DB	MDBR	2.6	2.8	2.5	2.3	2.0	1.9	1.7	1.9	2.0	2.1	2.3	2.6
			MCDBR	2.8	2.8	2.4	2.6	2.4	2.8	2.6	2.5	2.5	2.1	2.4	2.6
	MCWBR		3.4	3.2	2.6	2.3	1.8	2.0	2.0	2.1	2.1	2.4	3.1	3.7	
	5% WB	MCDBR	2.9	2.5	2.4	2.5	1.9	2.7	2.2	2.3	2.1	1.9	2.3	2.9	
		MCWBR	3.7	3.5	2.9	2.5	1.7	2.1	2.1	2.2	2.0	2.6	3.3	4.0	
Clear-Sky Solar Irradiance	taub	0.338	0.357	0.370	0.382	0.372	0.364	0.382	0.371	0.366	0.348	0.324	0.383		
	taud	2.062	2.181	2.261	2.301	2.401	2.464	2.432	2.462	2.423	2.303	2.126	2.284		
	Ebn at Noon	342	592	736	808	851	865	836	817	745	605	358	214		
	Edh at Noon	45	73	92	106	104	99	100	90	79	63	41	29		
All-Sky Solar Radiation	RadAvg	0.19	0.68	1.81	3.51	4.90	4.99	4.53	3.63	2.20	1.04	0.30	0.12		
	RadStd	0.03	0.08	0.19	0.27	0.35	0.54	0.41	0.34	0.17	0.12	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	+0.65	+0.62	N/A	N/A	N/A	-102	-134	N/A	N/A

Nomenclature: See separate page